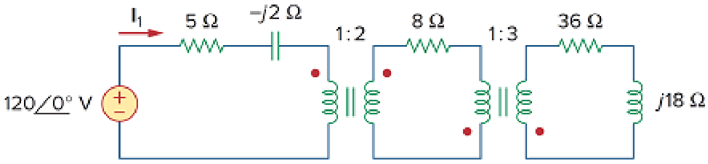
Chapter 13, Problem 51P

Problem

Use the concept of reflected impedance to find the input impedance and current **I**1 in Fig. 13.116.

Figure 13.116



Step-by-step solution

Step 1 of 3

Refer to Figure 13.116 in the textbook for a transformer circuit.

Use the reflected impedance concept to find the input impedance of the circuit.

Observe that there are two transformers in the circuit. First, reflect the secondary impedance of second transformer to the primary side.

The secondary impedance of the second transformer is,

$$Z_3 = (36 + j18)\ \Omega$$

The reflected impedance of the second transformer is,

$$Z_{32} = \frac{Z_3}{n_2^2}$$

Here, n_2 is the turns ratio of second transformer.

$$\begin{aligned} n_2 &= \frac{N_{22}}{N_{12}} \\ &= \frac{3}{1} \\ &= 3 \end{aligned}$$

Therefore,

$$\begin{aligned} Z_{32} &= \frac{Z_3}{n_2^2} \\ &= \frac{36 + j18}{3^2} \\ &= \frac{36 + j18}{9} \\ &= (4 + j2)\ \Omega \end{aligned}$$

Step 2 of 3

Calculate the secondary impedance of the first transformer.

$$\begin{aligned} Z_2 &= R_2 + Z_{32} \\ &= 8 + (4 + j2) \\ &= (12 + j2)\ \Omega \end{aligned}$$

The reflected impedance of the first transformer is,

$$Z_{21} = \frac{Z_2}{n_1^2}$$

Here, n_1 is the turns ratio of first transformer.

$$\begin{aligned} n_1 &= \frac{N_{21}}{N_{11}} \\ &= \frac{2}{1} \\ &= 2 \end{aligned}$$

Therefore,

$$\begin{aligned} Z_{21} &= \frac{Z_2}{n_1^2} \\ &= \frac{12 + j2}{2^2} \\ &= \frac{12 + j2}{4} \\ &= (3 + j0.5)\ \Omega \end{aligned}$$

Step 3 of 3

Calculate the input impedance of the circuit.

$$\begin{aligned} Z_{\text{in}} &= Z_{11} + Z_{21} \\ &= (5 - j2) + (3 + j0.5) \\ &= (8 - j1.5)\ \Omega \end{aligned}$$

Therefore, the input impedance Z_{in} is $\boxed{(8 - j1.5)\ \Omega}$.

Calculate the primary current $\mathbf{I}_1$ is,

$$\begin{aligned} \mathbf{I}_1 &= \frac{\mathbf{V}_S}{Z_{\text{in}}} \\ &= \frac{120\angle 0^\circ}{8 - j1.5} \\ &= \frac{120\angle 0^\circ}{8.14\angle -10.62^\circ} \\ &= 14.743\angle 10.62^\circ\ \text{A} \end{aligned}$$

Therefore, the primary current of first transformer $\mathbf{I}_1$ is $\boxed{14.743\angle 10.62^\circ\ \text{A}}$.